Empirical Methods in Finance

Class 5 - Multivariate Time Series Analysis

Florian Ielpo, f.ielpo@lombardodier.com

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Objectives of the lecture

When we turn to a multivariate framework, it is still possible to use time series tools, but at the same time to start thinking in terms of explanatory variables.

The simplest multivariate model is the **VAR model** (Vector AutoRegressive model), which allows modeling the interaction between a set of variables.

In this lecture, we will study:

1. VAR models
2. Causality relationships that can exist between variables
3. Impulse response analysis
4. Variance decomposition

From AR to VAR

Univariate ARMA models offer an interesting structure for analyze shocks:

$$x_t = \phi_0 + \phi_1 x_{t-1} + \sigma\epsilon_t \quad (1)$$

$$\epsilon_t \sim N(0,1). \quad (2)$$

The model can be split between:

- What we conditionally know: $\phi_0 + \phi_1 x_{t-1}$
- The unknown a.k.a. shocks: $\sigma\epsilon_t$.

In finance these shocks are the expression of randomness, in economics they can translate external stimulation coming from outside of the economy.
Multivariate equivalent would allow to study interactions between shocks and systems of variables.

From AR to VAR

What if faced with

$$\begin{cases} x_t = \phi_0 + \phi_1 x_{t-1} + \phi_2 y_{t-1} + \sigma \epsilon_t \\ y_t = \theta_0 + \theta_1 y_{t-1} + \theta_2 x_{t-1} + \gamma \nu_t \end{cases} \tag{3}$$

Adding the past of y_t to the dynamics of x_t makes it possible to let y_t's shocks affect x_t. Typical examples:

- Shocks to one asset's volatility contaminating other financial assets' volatility: price discovery and contagion effects.
- Simulation of macro shocks and how they spread through the economy.
- More recently: macro finance model, combining VAR-based Taylor rules and translating them into bonds/equity prices.

From AR to VAR

Matrix representation of the previous system:

$$\begin{bmatrix} x_2 & y_2 \\ x_3 & y_3 \\ \vdots & \vdots \\ x_n & y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_{n-1} & y_{n-1} \end{bmatrix} \begin{bmatrix} \phi_0 & \theta_0 \\ \phi_1 & \theta_1 \\ \phi_2 & \theta_2 \end{bmatrix} + \begin{bmatrix} \epsilon_2 & \nu_2 \\ \epsilon_3 & \nu_3 \\ \vdots & \vdots \\ \epsilon_n & \nu_n \end{bmatrix} \begin{bmatrix} \sigma & 0 \\ 0 & \gamma \end{bmatrix} \tag{4}$$

From which we obtain:

$$X_t = \Phi_0 + \Phi_1 X_{t-1} + \Sigma \epsilon_t \tag{5}$$

In this case:

- Σ is the matrix square root of the covariance matrix of shocks.
- This matrix is diagonal: shocks at time t affecting x_t are not affecting y_t at time t.
- It contains only one lag, meaning the structure of persistence of shocks is quite simple.

Weak Stationarity and Cross-Correlation Matrices

Notations

An n-dimensional time series $r_t = (r_{1,t}, ..., r_{n,t})'$ is **weakly stationary** if its mean vector and covariance matrix are time invariant.
They are defined as

- $\mu = E[r_t]$
- $\Gamma_0 = E[(r_t - \mu)(r_t - \mu)']$

The mean $\mu = (\mu_1, ..., \mu_n)'$ is an $(n \times 1)$ vector of unconditional expectations of the $r_{i,t}$.

The covariance matrix $\Gamma_0 = [\Gamma_{ij}(0)]$ is an $(n \times n)$ matrix.

The i^{th} diagonal element is the variance of $r_{i,t}$ and the $(i,j)^{th}$ element is the covariance between $r_{i,t}$ and $r_{j,t}$.

Cross-correlation Matrices

Let D be a $(n \times n)$ diagonal matrix of variances of $r_{i,t}$ for $i = 1, ..., n$s:

$$D = \begin{pmatrix} \Gamma_{11}(0) & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \Gamma_{nn}(0) \end{pmatrix} \tag{6}$$

The **lag-zero cross-correlation matrix** of r_t is

$$\rho_0 = [\rho_{ij}(0)] = D^{-1/2}\Gamma_0 D^{-1/2} \tag{7}$$

with $(i,j)^{th}$ element $\rho_{ij}(0) = \frac{\Gamma_{ij}(0)}{\sqrt{\Gamma_{ii}(0)\Gamma_{jj}(0)}} = \frac{Cov[r_{i,t}, r_{j,t}]}{\sqrt{V[r_{i,t}]V[r_{j,t}]}}$.

Obviously, we have $\rho_{ij}(0) = \rho_{ji}(0)$, $-1 \leq \rho_{ij}(0) \leq 1$ and $\rho_{ii}(0) = 1$.

Cross-correlation Matrices

The **lag-*k* cross-covariance matrix** of r_t describes the lead-lag relationship:

$$\Gamma_k = [\Gamma_{ij}(k)] = E[(r_t - \mu)(r_{t-k} - \mu)'] \tag{8}$$

Therefore, the $(i,j)^{th}$ element of Γ_k is the covariance between $r_{i,t}$ and $r_{j,t-k}$.

Weak Stationarity and Cross-Correlation Matrices

The **lag-k cross-correlation matrix** of r_t is

$$\rho_k = [\rho_{ij}(k)] = D^{-1/2}\Gamma_k D^{-1/2} \tag{9}$$

with $(i,j)^{th}$ element $\rho_{ij}(k) = \frac{\Gamma_{ij}(k)}{\sqrt{\Gamma_{ii}(0)\Gamma_{jj}(0)}} = \frac{Cov[r_{i,t}, r_{j,t-k}]}{\sqrt{V[r_{i,t}]V[r_{j,t}]}}$
which is the correlation coefficient between $r_{i,t}$ and $r_{j,t-k}$.

$\rho_{ij}(k)$ measures the linear dependence of $r_{i,t}$ on $r_{j,t-k}$. If $\rho_{ij}(k) \neq 0$, we say that the series $r_{j,t}$ leads the series $r_{i,t}$ at lag k.

Cross-correlation Matrices

Similarly, $\rho_{ji}(k)$ measures the linear dependence of $r_{j,t}$ on $r_{i,t-k}$. If $\rho_{ji}(k) \neq 0$, we say that the series $r_{i,t}$ leads the series $r_{j,t}$ at lag k.
In general $\rho_{ji}(k) \neq \rho_{ij}(k)$ for $i \neq j$ because the two correlation coefficients measure different linear relationships between $r_{i,t}$ and $r_{j,t}$. So, Γ_k and ρ_k are **not symmetric**.

Under weak stationarity, we have $\Gamma_{ij}(k) = \Gamma_{ji}(-k)$.

Equivalently, $\Gamma_k = \Gamma'_{-k}$ and $\rho_k = \rho'_{-k}$.

So, it is sufficient in practice to compute the cross-correlation matrices ρ_k for $k \geq 0$.

Sample Cross-correlation Matrices

Now, let $\{r_t\}_{t=1}^{T}$ be an n-dimensional time series. The cross-covariance matrix Γ_k is estimated by

$\hat{\Gamma}_k = \frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})(r_{t-k} - \bar{r})' \qquad k \geq 0$ with $\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t$

The cross-correlation matrix ρ_k is estimated by

$$\hat{\rho}_k = \hat{D}^{-1/2}\hat{\Gamma}_k\hat{D}^{-1/2} k \geq 0 \tag{10}$$

where $\hat{D}$ is the $(n \times n)$ diagonal matrix of the sample variances.

The finite-sample distribution is complex in presence of conditional heteroskedasticity and non-normality.

VAR(1) Model

A simple model for modeling the joint dynamics of asset returns is the **Vector AutoRegressive (VAR) model**.

A multivariate time series r_t is a VAR(1) process if

$$r_t = \Phi_0 + \Phi_1 r_{t-1} + \varepsilon_t \tag{11}$$

where Φ_0 is an $(n \times 1)$ vector, Φ_1 is an $(n \times n)$ matrix and ε_t is a sequence of serially uncorrelated random vectors with mean zero and covariance matrix Σ.

In general, we assume that Σ is positive definite and ε_t is multivariate normal.

In the bivariate case, the VAR(1) model can be written as

$$\begin{aligned} r_{1,t} &= \varphi_{10} + \varphi_{11} r_{1,t-1} + \varphi_{12} r_{2,t-1} + \varepsilon_{1,t} \\ r_{2,t} &= \varphi_{20} + \varphi_{21} r_{1,t-1} + \varphi_{22} r_{2,t-1} + \varepsilon_{2,t} \end{aligned} \tag{12}$$

Stationarity Condition and Moments

Assume that the VAR(1) model is **weakly stationary**.

Since $\mathbb{E}[\varepsilon_t] = 0$, we have $\mathbb{E}[r_t] = \Phi_0 + \Phi_1 \mathbb{E}[r_{t-1}]$.

Since $\mathbb{E}[r_t]$ is time invariant, we have $\mathbb{E}[r_t] = \mu = (I_n - \Phi_1)^{-1}\Phi_0$ provided the matrix $(I_n - \Phi_1)$ is nonsingular, with I_n the $(n \times n)$ identity matrix.

Therefore, the VAR(1) model can then be written as

$$(r_t - \mu) = \Phi_1 (r_{t-1} - \mu) + \varepsilon_t \tag{13}$$
$$\text{or} \tilde{r}_t = \Phi_1 \tilde{r}_{t-1} + \varepsilon_t, \tag{14}$$

where $\tilde{r}_t = (r_t - \mu)$ is the mean-corrected time series.
By repeating substitution, we have the **Wold decomposition:**

$$\tilde{r}_t = \varepsilon_t + \Phi_1 \varepsilon_{t-1} + \Phi_1^2 \varepsilon_{t-2} + \Phi_1^3 \varepsilon_{t-3} + ... \tag{15}$$

VAR(1) Model - Stationarity Condition and Moments

This expression provides several characteristics of the VAR(1) process.

1. Since ε_t is serially uncorrelated, it follows that

$$Cov[r_t, \varepsilon_t] = \Sigma \tag{16}$$

$$Cov[r_{t-k}, \varepsilon_t] = 0 \forall k > 0 \tag{17}$$

2. Φ_1^k must converge to zero as $k \to \infty$, or the n eigenvalues of Φ_1 must be less than 1 in modulus. This is the **necessary and sufficient condition for weak stationarity of r_t** (provided that the covariance matrix Σ exists).
3. Also

$$V[r_t] = \Gamma_0 = \Sigma + \Phi_1 \Sigma \Phi_1' + \Phi_1^2 \Sigma (\Phi_1^2)' + ... = \sum_{i=0}^{\infty} \Phi_1^i \Sigma (\Phi_1^i)' \tag{18}$$

$$\text{with } \Phi_1^0 = I_n \tag{19}$$

VAR(p) Models

The generalization of the VAR(1) model to the VAR(p) model is straightforward:

$$\begin{aligned} r_t &= \Phi_0 + \Phi_1 r_{t-1} + ... + \Phi_p r_{t-p} + \varepsilon_t \\ (I_n - \Phi_1 L - ... - \Phi_p L^p) r_t &= \Phi(L) r_t = \Phi_0 + \varepsilon_t \end{aligned} \tag{20}$$

where $\Phi(L)$ is a matrix polynomial. If r_t is **weakly stationary**, we have

$$\mathbb{E}[r_t] = (I_n - \Phi_1 - ... - \Phi_p)^{-1} \Phi_0 = \Phi(1)^{-1} \Phi_0 \tag{21}$$

provided the matrix $\Phi(1)$ is non-singular. Also:

- $Cov[r_t, \varepsilon_t] = \Sigma$
- $Cov[r_{t-k}, \varepsilon_t] = 0, \forall k > 0$
- $\Gamma_k = \Phi_1 \Gamma_{k-1} + ... + \Phi_p \Gamma_{k-p} \forall k > 0$

VAR(p) Model: Estimation

The objective is to estimate the parameters of the (centered) n-dimensional VAR(p):

$$r_t = \Phi_1 r_{t-1} + ... + \Phi_p r_{t-p} + \varepsilon_t \tag{22}$$

In a matrix form, it becomes:

$$\underset{(T,N)}{r} = \underset{(T,N)}{r_{-1}} \underset{(N,N)}{\Phi_1} + ... + r_{-p}\Phi_p + \varepsilon \tag{23}$$

$$\Leftrightarrow \underset{(T,N)}{r} = \underset{(T,Np)}{R} \underset{(Np,N)}{\Phi} + \varepsilon \tag{24}$$

with $\underset{(T,N)}{r} = \begin{pmatrix} r_{1,1} & \cdots & r_{N,1} \\ \vdots & \ddots & \vdots \\ r_{1,T} & \cdots & r_{N,t} \end{pmatrix}$, $\underset{(T,Np)}{R} = (r_{-1}, \cdots, r_{-p})$ and $\underset{(Np,N)}{\Phi} = \begin{pmatrix} \Phi_1 \\ \vdots \\ \Phi_p \end{pmatrix}$

VAR(p) Model: Estimation

With this form, we can use the usual formula of the OLS regression:

$$\underset{(Np,N)}{\hat{\Phi}} = (R'R)^{-1}R'r \tag{25}$$

The residual matrix is defined as $\hat{\varepsilon} = r - R\,\hat{\Phi}$.
The residual covariance matrix is asymptotically: $\hat{\Sigma} = \frac{\hat{\varepsilon}'\hat{\varepsilon}}{T}$.
If we denote $\hat{Q} = \frac{R'R}{T}$, the estimates' covariance matrix becomes:

$$\hat{V}(\hat{\Phi}) = \hat{\Sigma} \otimes Q^{-1}. \tag{26}$$

VAR(p) Model: Estimation

Estimation can be done also by maximum likelihood:

$$\log L = \sum_{t=2}^{n} \log f\left(X_t\right), \tag{27}$$

with $f(.)$ here being the density of a Gaussian multivariate distribution.

The distribution function of a Multivariate Gaussian random variable:
If $X_t = [x_1, x_2, \ldots, x_k]$ follows a multivariate Gaussian random variable with expectation Γ and covariance matrix Σ, its distribution function writes

$$f(X_t) = \frac{1}{(2\pi)^{\frac{k}{2}}|\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(X-t-\Gamma)^{\top}\Sigma^{-1}(X_t-\Gamma)} \tag{28}$$

Several remarks:

- $|\Sigma|$ is the determinant of Σ, meaning:

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc. \tag{29}$$

- The density is a scalar, even if computed from multivariate matrices.
- The log density writes:

$$\log f(X_t) = -\frac{k}{2}\log(2\pi) - \frac{1}{2}\det(\Sigma) - \frac{1}{2}(X-\Gamma)^\top\Sigma^{-1}(X-\Gamma) \tag{30}$$

The log likelohood writes:

$$\log L = -\frac{(n - \max(p,q))k}{2}\log(2\pi) \tag{31}$$
$$-\frac{n}{2}\det(\Sigma) \tag{32}$$
$$-\frac{1}{2}\sum_{t=\max(p,q)+1}^{n}(X_t - \mathbb{E}_t[X_t])^\top \Sigma^{-1}(X_t - \mathbb{E}_t[X_t]) \tag{33}$$

with $\mathbb{E}[X_t] = \Phi_0 + \sum_{i=1}^{p}\Phi_i X_{t-i} + \sum_{j=1}^{q}\Theta_j \Sigma \epsilon_{t-j}$.
Not much more complex than in the ARMA case BUT dimensionality will prove costly.
A VAR(1) model with three variables will require estimating: 3 parameters for Φ_0, 9 parameters for Φ_1 and 6 parameters for Σ.
This is in total 18 parameters to be estimated. It grows rapidly as a function of k and p. With $k = 4$, the number of parameters is 30. With $k = 3, p = 2$ this number is 27.

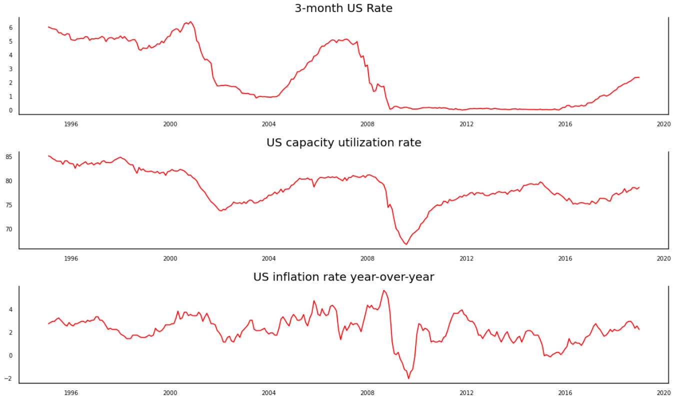
Test model: back on the Taylor Rule.

Dataset:

1. 3-month US Rate as a proxy for the Fed's target rate
2. US capacity utilization rate
3. US inflation rate year-over-year.

Time period: 31.01.1995 - 31.12.2018.

VARMA Models



Determining the Order of a VAR Model

The parameters of a VAR(p) model can be estimated using either the OLS and ML methods. The two methods are asymptotically equivalent. Under some regularity conditions, the estimates are asymptotically normal.

As for the univariate AR(p) process, we can determine the order of the VAR(p) process using a **sequential testing procedure**. Consider the n-dimensional VAR(p):

$$r_t = \Phi_0 + \Phi_1 r_{t-1} + ... + \Phi_p r_{t-p} + \varepsilon_t. \tag{34}$$

VAR(p) Model

Let $\hat{\Phi}_j^{(p)}$ be the OLS estimate of Φ_j where the superscript (p) means that the estimates are for a VAR(p). Then the residual is: $\hat{\varepsilon}_t^{(p)} = r_t - \hat{\Phi}_0^{(p)} - \hat{\Phi}_1^{(p)} r_{t-1} - ... - \hat{\Phi}_p^{(p)} r_{t-p}$.

If $p = 0$, then the residual simplifies to $\hat{\varepsilon}_t^{(0)} = r_t - \bar{r}$.

The residual covariance matrix is $\hat{\Sigma}^{(p)} = \frac{1}{T-np-1} \sum_{t=p+1}^{T} \hat{\varepsilon}_t^{(p)} (\hat{\varepsilon}_t^{(p)})'$

Determining the Order of a VAR Model

Now, assume that we want to test the null hypothesis $H_0 : \Phi_p = 0$ versus the alternative hypothesis $H_a : \Phi_p \neq 0$.
The **likelihood ratio test statistic** is

$$LR(p) = 2\left(\log(L^{(p)}) - \log(L^{(p-1)})\right) = (T - np - 1)\left[\log\left(\left|\hat{\Sigma}^{(p-1)}\right|\right) - \log\left(\left|\hat{\Sigma}^{(p)}\right|\right)\right] \tag{35}$$

Under some regularity conditions, the LR statistic is asymptotically a $\chi^2(n^2)$.

VAR(p) Model

Alternatively, we may use **information criteria** to select the optimal order of the VAR(p) model. The AIC and BIC criteria are defined as, for $p \in 0, ..., p_{\text{max}}$.

$AIC(p) = \log\left(\left|\hat{\Sigma}^{(p)}\right|\right) + \frac{2n^2p}{T}$ and $BIC(p) = \log\left(\left|\hat{\Sigma}^{(p)}\right|\right) + \frac{n^2p}{T}\log(T)$

Then, the optimal order p^* is selected such that one these information criterion is minimized.

Determining the Order of a VAR Model

Two other criterions:

1. The Final Prediction Error (FPE):

$$FPE(p) = \left(\left|\hat{\Sigma}^{(p)}\right|\right) + \left(\frac{T + np + 1}{T - np - 1}\right)^n . \tag{36}$$

2. The Hannan-Quinn Criterion (HQ):

$$HQ(p) = \log\left(\left|\hat{\Sigma}^{(p)}\right|\right) + 2\frac{\log\log T}{T} pn^2 . \tag{37}$$

Here again, the "best" model is the model that minimizes one of these information criterion.

In practice, people rely on a breadth of these criterion. The AIC is potentially the least used and the HQ the most used.

VARMA Models

```
   ...: x.summary()
Lag Order = 1
AIC :   -6.498649593901928
BIC :   -6.345639884532188
FPE :   0.0015054789703375755
HQIC:   -6.4373256295772565

Lag Order = 2
AIC :   -6.774898956677228
BIC :   -6.506452005533113
FPE :   0.0011421193975505134
HQIC:   -6.667297382585719

Lag Order = 3
AIC :   -6.917866328645517
BIC :   -6.53339378335408
FPE :   0.0009900254892843171
HQIC:   -6.763740852641638

Lag Order = 4
AIC :   -6.958604398816028
BIC :   -6.457512866110792
FPE :   0.0009506046212708003
HQIC:   -6.757706616151132
```

VARMA Models

Results for equation 3-month US Rate

	coefficient	std. error	t-stat	prob
const	-0.811606	0.380103	-2.135	0.033
L1.3-month US Rate	0.981944	0.008410	116.755	0.000
L1.US capacity utilization rate	0.011483	0.005108	2.248	0.025
L1.US inflation rate year-over-year	-0.026029	0.011135	-2.337	0.019

Results for equation US capacity utilization rate

	coefficient	std. error	t-stat	prob
const	1.117886	0.992541	1.126	0.260
L1.3-month US Rate	0.002110	0.021961	0.096	0.923
L1.US capacity utilization rate	0.985625	0.013338	73.895	0.000
L1.US inflation rate year-over-year	-0.008763	0.029077	-0.301	0.763

Results for equation US inflation rate year-over-year

	coefficient	std. error	t-stat	prob
const	-0.351461	0.805038	-0.437	0.662
L1.3-month US Rate	0.014505	0.017813	0.814	0.415
L1.US capacity utilization rate	0.006436	0.010818	0.595	0.552
L1.US inflation rate year-over-year	0.915510	0.023584	38.819	0.000

Now if we need to impose a constaint:

```
####
# Estimation of a constrained VAR
import numpy.matlib
from scipy.stats import multivariate_normal

def ML_VAR(theta,X):
    Phi1=[[theta[0], theta[1], theta[2]],
    [theta[3], theta[4], theta[5]],
    [theta[6], theta[7], theta[8]]]
    Phi0=[[theta[9], theta[10], theta[11]]]

    Y=X.iloc[1:np.size(X,0),:];
    Z=X.iloc[0:(np.size(X,0)-1),:]
    Phi0_temp=np.matlib.repmat(Phi0,np.size(Z,0),1)
    temp=np.matmul(Z,Phi1)-Phi0_temp
    res=np.subtract(Y,temp)
    loglik=multivariate_normal.logpdf(res, mean=[0,0,0], cov=np.identity(3))
    loglik=-np.sum(loglik)
    return loglik
```

Now if we need to impose a constaint:

```
Phi1=model_fitted.coefs;
Phi0=model_fitted.coefs_exog;
theta=[*Phi1.flatten(),*Phi0.flatten()]

from scipy.optimize import minimize

estimation_output = minimize(ML_VAR, theta, method='BFGS', args=(Macro_series));
estimated_para=estimation_output.x;
standard_deviation=np.power(np.diag(estimation_output.hess_inv),.5);
t_stat=estimated_para/standard_deviation
```

VARMA Models

```
In [221]: estimated_para
Out[221]:
array([ 0.98195488,  0.00212096,  0.01451694,  0.01147413,  0.98561594,
        0.00642592, -0.02602356, -0.00875731,  0.91551484,  0.81091284,
       -1.11858472,  0.35074358])

In [222]: standard_deviation
Out[222]:
array([0.03082015, 0.10023432, 0.03078478, 0.0028836 , 0.0292109 ,
       0.00348349, 0.05671925, 0.1504932 , 0.05633414, 0.16968513,
       2.32747764, 0.24948763])

In [223]: t_stat
Out[223]:
array([ 3.18608047e+01,  2.11600067e-02,  4.71562062e-01,  3.97909434e+00,
        3.37413756e+01,  1.84467954e+00, -4.58813552e-01, -5.81907068e-02,
        1.62515098e+01,  4.77892684e+00, -4.80599559e-01,  1.40585557e+00])
```

VMA(q) Model

A vector moving-average model of order q, or VMA(q), is in the form

$$r_t = \mu + \varepsilon_t - \Theta_1 \varepsilon_{t-1} - \ldots - \Theta_q \varepsilon_{t-q} = \mu + \Theta(L)\varepsilon_t \tag{38}$$

where μ is an $(n \times 1)$ vector, Θ_i are $(n \times n)$ matrices and $\Theta(L) = I_n - \Theta_1 L - \ldots - \Theta_q L^q$. VMA($q$) processes are **weakly stationary** if the covariance matrix Σ of ε_t exists.

Using the fact that ε_t has no serial correlation, we have

- $\mu = \mathbb{E}[r_t]$
- $Cov(r_t, \varepsilon_t) = \Sigma$
- $\Gamma_0 = \Sigma + \Theta_1 \Sigma \Theta_1^{'} + \ldots + \Theta_q \Sigma \Theta_q^{'}$
- $\Gamma_k = \sum_{i=k}^{q} \Theta_i \Sigma \Theta_{i-k+1}^{'}$ if $1 \leq k \leq q$ with $\Theta_0 = I_n$
- $\Gamma_k = 0$ and $\rho_k = 0$ for $k > q$, so that the sample cross-correlation matrices can be used to identify the order of a VMA process.

Consider the bivariate MA(1) model

$$\begin{pmatrix} r_{1,t} \\ r_{2,t} \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix} + \begin{pmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-1} \\ \varepsilon_{2,t-1} \end{pmatrix} \tag{39}$$

meaning that the return series at time t only depend on the current and past shocks. This is therefore a finite-memory model.

The off-diagonal elements of Θ thus show the dynamic dependence between the component series.

VMA(q) Model

We have therefore the following relationships:

- $r_{1,t}$ and $r_{2,t}$ are uncoupled series if $\theta_{12} = \theta_{21} = 0$
- There is a unidirectional dynamic relationship from $r_{1,t}$ to $r_{2,t}$ if $\theta_{12} = 0$ but $\theta_{21} \neq 0$.
- The opposite unidirectional dynamic relationship holds if $\theta_{21} = 0$ but $\theta_{12} \neq 0$.
- There is a feedback relationship between $r_{1,t}$ and $r_{2,t}$ if $\theta_{12} \neq 0$ and $\theta_{21} \neq 0$.

Estimation

Unlike the VAR model, the estimation of VMA models is much more involved, because there is no analytical solution and only numerical methods are available.

For the ML approach, there are two methods:

1. The **conditional ML** method that assumes that $\varepsilon_t = 0$ for $t \leq 0$.
2. The **exact ML** method that treats ε_t for $t \leq 0$ as additional parameters.

When the number of observations T is large, the two methods give the same estimates. Since T is generally large in finance, we focus on the conditional ML method.

Estimation

We consider a VMA(1) model and assume that $\varepsilon_0 = 0$. Under such assumption and rewriting the model as $\varepsilon_t = r_t - \mu - \Theta_1 \varepsilon_{t-1}$, we compute the shock ε_t recursively as

$$\varepsilon_1 = r_1 - \mu \tag{40}$$

$$\varepsilon_2 = r_2 - \mu - \Theta_1 \varepsilon_1 \tag{41}$$

$$\cdots \tag{42}$$

So, the likelihood function of the data becomes

$$f(r_1, ..., r_T | \mu, \Theta_1, \Sigma) = \prod_{\left(-\frac{1}{2}\varepsilon_t' \Sigma^{-1} \varepsilon_t\right)} \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}\epsilon_t^\top \Sigma^{-1} \epsilon_t\right) \tag{43}$$

This expression can be maximized to obtain the parameter estimates.

VARMA(p,q) Model

A vector autoregressive moving-average model of order p and q, or VARMA(p,q), is written as

$$\begin{aligned} r_t &= \mu + \textstyle\sum_{i=1}^{p} \Phi_i r_{t-i} + \varepsilon_t + \sum_{i=1}^{q} \Theta_i \varepsilon_{t-i} \\ \Phi(L) r_t &= \mu + \Theta(L) \varepsilon_t \end{aligned} \tag{44}$$

where Φ_i and Θ_i are $(n \times n)$ matrices such that $\Phi(L) = I_n - \Phi_1 L - ... - \Phi_p L^p$ and$\Theta(L) = I_n - \Theta_1 L - ... - \Theta_q L^q$, and μ is an $(n \times 1)$ vector.

The generalization of ARMA models to the multivariate case encounters some new issues that do not occur in developing VAR and VMA models.

VARMA(p,q) Model

The main problem with VARMA is **identifiability**. Unlike the univariate ARMA models, VARMA models may not be uniquely defined.
For instance the VMA(1) model

$$\begin{pmatrix} r_{1,t} \\ r_{2,t} \end{pmatrix} = \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix} + \begin{pmatrix} 0 & -2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-1} \\ \varepsilon_{2,t-1} \end{pmatrix} \tag{45}$$

is identical to the VAR(1) model

$$\begin{pmatrix} r_{1,t} \\ r_{2,t} \end{pmatrix} - \begin{pmatrix} 0 & -2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} r_{1,t-1} \\ r_{2,t-1} \end{pmatrix} = \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix} \tag{46}$$

VARMA(p,q) Model

The equivalence of the two models is easy to show since we have

VMA model	VAR model
$r_{1,t} = \varepsilon_{1,t} - 2\varepsilon_{2,t-1}$	$r_{1,t} + 2r_{2,t-1} = \varepsilon_{1,t}$
$r_{2,t} = \varepsilon_{2,t}$	$r_{2,t} = \varepsilon_{2,t}$

In general VAR and VMA models are sufficient in most financial applications to capture the statistical features of the data.

Test for Granger Causality

Granger causality is defined in terms of linear predictions.

Definition: Y_t does not **Granger cause** X_t iif the best linear prediction of X_t given X_{t-1} and Y_{t-1} does not depend on Y_{t-1}. More formally, let $\Omega_{t-1}(X)$ be the information set containing all the lags of X: $\Omega_{t-1}(X) = (X_\tau, \tau \leq t-1)$. Now, we say Y_t does not Granger cause X_t iif

$$MSE[X_t|\Omega_{t-1}(X), \Omega_{t-1}(Y)] = MSE[X_t|\Omega_{t-1}(X)] \qquad \forall t \tag{47}$$

where MSE is the mean squared error

$$MSE[X_t|\Omega_{t-1}(X)] = \frac{1}{T}\sum_{t=1}^{T} \hat{\varepsilon}^2_{X,t} \tag{48}$$

With $\hat{\varepsilon}_{X,t} = X_t - EL[X_t|\Omega_{t-1}(X)]$ where EL denotes linear regression.

Test for Granger Causality

The test of Granger causality is very simple to implement in a VAR model (see Geweke, 1982). Consider the three following representations:

- **The VAR(p) model:**

$$\begin{aligned} Y_t &= \Phi_{11}(L)Y_t + \Phi_{12}(L)X_t + \varepsilon_{1,t} \\ X_t &= \Phi_{22}(L)X_t + \Phi_{21}(L)Y_t + \varepsilon_{2,t} \end{aligned} \tag{49}$$

  with $V\begin{bmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{bmatrix} = \Sigma$

- **The joint dynamics** of Y_t and X_t assuming no cross-correlation between the error terms:
  $Y_t = M_{11}(L)Y_t + M_{12}(L)X_t + \eta_{1,t}$ with $V[\eta_{1,t}] = \Sigma_{12}$
  $X_t = M_{22}(L)X_t + M_{21}(L)Y_t + \eta_{2,t}$ with $V[\eta_{2,t}] = \Sigma_{21}$
- **The marginal dynamics** of Y_t and X_t as function of their own lags only: $Y_t = m_1(L)Y_t + \nu_{1,t}$ with $V[\nu_{1,t}] = \Sigma_1$
  $X_t = m_2(L)X_t + \nu_{2,t}$ with $V[\nu_{2,t}] = \Sigma_2$

Test for Granger Causality - The non-causality hypotheses and the causality measures

- The test statistic for the null hypothesis that Y does not Granger cause X ($H_{Y \to X}$) is

$$C_{Y \to X} = \log \left(\frac{|\Sigma_2|}{|\Sigma_{21}|} \right) \tag{50}$$

- The test statistic for the null hypothesis that X does not Granger cause Y ($H_{X \to Y}$) is

$$C_{X \to Y} = \log \left(\frac{|\Sigma_1|}{|\Sigma_{12}|} \right) \tag{51}$$

- The test statistic for the null hypothesis that there is no instantaneous relationship between X and Y ($H_{Y \leftrightarrow X}$) is

$$C_{X \leftrightarrow Y} = \log \left(\frac{|\Sigma_{12}||\Sigma_{21}|}{|\Sigma|} \right). \tag{52}$$

Test for Granger Causality - The non-causality hypotheses and the causality measures

The test statistic for the null hypothesis that there is no interdependence between Y and X is

$$C_{Y,X} = \log\left(\frac{|\Sigma_1||\Sigma_2|}{|\Sigma|}\right) = C_{Y\to X} + C_{X\to Y} + C_{X\leftrightarrow Y}. \tag{53}$$

Empirical counterparts to the covariance elements need to be used to obtain the test statistics.

Then under the null hypothesis, the asymptotic distributions for the four measures of causality are (Geweke, 1982):

$$\begin{aligned} T \times C_{Y\to X} &\overset{a}{\sim} \chi^2(p) \\ T \times C_{Y\leftrightarrow X} &\overset{a}{\sim} \chi^2(1) \end{aligned} \tag{54}$$

$$\begin{aligned} T \times C_{X\to Y} &\overset{a}{\sim} \chi^2(p) \\ T \times C_{Y,X} &\overset{a}{\sim} \chi^2(2p+1) \end{aligned} \tag{55}$$

Impulse Responses

Impulse responses are based on the Wold decomposition of a multivariate model:

$$r_t = \varepsilon_t + \Psi_1 \varepsilon_{t-1} + \Psi_2 \varepsilon_{t-2} + ... = \Psi(L)\varepsilon_t \tag{56}$$

where the matrix Ψ_s has the interpretation $\Psi_s = \frac{\partial r_{t+s}}{\partial \varepsilon_t'}$

Impulse Responses

The element (i,j) of matrix Ψ_s is the impact of a one-unit increase in the jth variable's innovation at date t ($\varepsilon_{j,t}$) for the value of the ith variable at time $t+s$ ($r_{i,t+s}$), when all other innovations at all dates are constant.

If we assume that the first element of ε_t is changed by δ_1, ..., the nth element of ε_t by δ_n, then the combined effect of these changes on the value of the vector r_{t+s} is given by

$$\Delta r_{t+s} = \frac{\partial r_{t+s}}{\partial \varepsilon_{1,t}}\delta_1 + \frac{\partial r_{t+s}}{\partial \varepsilon_{2,t}}\delta_2 + ... + \frac{\partial r_{t+s}}{\partial \varepsilon_{n,t}}\delta_n = \Psi_s \delta \tag{57}$$

A plot of element (i,j) of Ψ_s as a function of s is the **impulse response function**.

Impulse Responses: Case of VAR(1) Model

Assume a VAR(1) model

$$r_t = \Phi_1 r_{t-1} + \varepsilon_t \tag{58}$$

where Φ_1 is an $(n \times n)$ matrix and $\varepsilon_t \sim (0, \Sigma)$.

The relation between r_t and a past shocks is given by

$$r_t = \Phi_1(\Phi_1 r_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \varepsilon_t + \Phi_1 \varepsilon_{t-1} + \Phi_1^2 \varepsilon_{t-2} + ... \tag{59}$$

Therefore, the impact of a one-unit increase in innovations at date t (ε_t) for the value of the r_{t+s} is Φ_1^s.

Remark: The impulse response is another kind of causality measure, although it is different from Granger causality.

Consider the following model

$$\begin{pmatrix} r_{1,t} \\ r_{2,t} \\ r_{3,t} \end{pmatrix} = \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} r_{1,t-1} \\ r_{2,t-1} \\ r_{3,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \end{pmatrix} \tag{60}$$

or $r_t = \breve{\ }_1 r_{t-1} + \varepsilon_t$.

Assume a unit shock in the first variable at time $t = 0$:

$$r_0 = \begin{pmatrix} r_{1,0} \\ r_{2,0} \\ r_{3,0} \end{pmatrix} = \begin{pmatrix} \varepsilon_{1,0} \\ \varepsilon_{2,0} \\ \varepsilon_{3,0} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \tag{61}$$

Then at time $t = 1$, we have:

$$r_1 = \begin{pmatrix} r_{1,1} \\ r_{2,1} \\ r_{3,1} \end{pmatrix} = \breve{}_1 r_0 = \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.1 \\ 0 \end{pmatrix} \tag{62}$$

At time $t = 2$, we have:

$$r_2 = \begin{pmatrix} r_{1,2} \\ r_{2,2} \\ r_{3,2} \end{pmatrix} = \breve{}_1 r_1 = \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 0.5 \\ 0.1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.25 \\ 0.06 \\ 0.02 \end{pmatrix} \tag{63}$$

Impulse Responses: Example

In the previous cases, we have assumed a shock in only one variable at a time. This assumption is reasonable if shocks are independent.

When shocks are not independent, which is likely as $\varepsilon_t \sim (0, \Sigma)$, we need to orthogonalize them. We consider the following Cholesky decomposition: $\Sigma = PP'$ where P is a lower triangular matrix. Then we can transform

$$r_t = \varepsilon_t + \Psi_1 \varepsilon_{t-1} + \Psi_2 \varepsilon_{t-2} + ... = \Psi(L)\varepsilon_t \tag{64}$$

as follows

$$r_t = \Gamma_0 w_t + \Gamma_1 w_{t-1} + \Gamma_2 w_{t-2} + ... = \Gamma(L) w_t \tag{65}$$

where $\Gamma_s = \Psi_s P$, $w_t = P^{-1}\varepsilon_t$ and $V[w_t] = I_n$.

Matrices Γ_s correspond to **orthogonalized impulse responses**.

Impulse Response Functions

Study of the impact of a shock in one variable over a set of variables.
Shocks are captured through ϵ_t and their instantaneous impact through $\sqrt{\Sigma}$.

Assume a VAR(1) with two variables:

$$X_t = \Phi_0 + \Phi_1 X_{t-1} + \sqrt{\Sigma}\epsilon_t, \tag{66}$$

with $\epsilon_t = \begin{pmatrix} \epsilon_t^1 & \epsilon_t^2 \end{pmatrix}$.
Assume Σ has the following form:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \tag{67}$$

With that specification, shocks are instantaneously *orthogonal*, meaning shocks in variable 1 do no influence the shocks in variable 2.

Now assume Σ is as follows:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix} \tag{68}$$

Assume we have a unit shock on variable 1 and that

$$\sqrt{\Sigma} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}. \tag{69}$$

The actual shock on variable 2 writes:

$$x_t^{(2)} = \Phi_0^{(2)} + \Phi_1^{(12)} x_{t-1}^{(1)} + \Phi_1^{(22)} x_{t-1}^{(2)} + \underbrace{b\epsilon_t^{(1)} + c\epsilon_t^{(2)}}_{\text{Variable 2 shock}} \tag{70}$$

Shocks for one variable are therefore a cocktail of the orthogonal shocks contained in ϵ_t. How can we simulate shocks and their dynamic impact of the economy?

Impulse Response Functions

VAR models are *stationary* models.
Their Wold decomposition writes:

$$X_t = \mu + \Psi_0 \epsilon_t + \Psi_1 \epsilon_{t-1} + \Psi_2 \epsilon_{t-2} + \dots \tag{71}$$

With this representation in mind, it is obvious that

$$\frac{\partial X_t}{\partial \epsilon_{t-i}} = \Psi_i. \tag{72}$$

Ψ_i measure the influence of ϵ_{t-i} on X_t.

We need to orthogonalize the shocks. Any definite positive matrix Σ can be decomposed as follows

$$\Sigma = PP^\top, \tag{73}$$

where

$$P = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}. \tag{74}$$

P is an invertible matrix.

Impulse Response Functions

We can re-write all shocks as

$$u_t = P^{-1}\epsilon_t. \tag{75}$$

With that in mind:

$$\mathbb{E}[u_t] = \mathbb{E}[P^{-1}\epsilon_t] = P^{-1}\mathbb{E}[\epsilon_t] = 0 \tag{76}$$

$$\mathbb{V}[u_t] = \mathbb{E}\left[u_t u_t^\top\right] = \mathbb{E}\left[P^{-1}\epsilon_t\epsilon_t^\top(P^{-1})^\top\right] = P^{-1}PP^\top(P^\top)^{-1} = I. \tag{77}$$

u_t is therefore an orthogonalized set of shocks. As

$$\epsilon_t = Pu_t, \tag{78}$$

we can rewrite the Wold decomposition as follows:

$$X_t = \mu + \Psi_0 Pu_t + \Psi_1 Pu_{t-1} + \Psi_2 Pu_{t-2} + \dots \tag{79}$$

Shocks in u_t are now orthogonal.

The system of shocks can be re-written as follows:

$$\begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \\ \epsilon_{3t} \\ \vdots \end{bmatrix} = \begin{bmatrix} P_{11} & 0 & 0 & \dots \\ P_{21} & P_{22} & 0 & \dots \\ P_{31} & P_{32} & P_{33} & \dots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \\ \vdots \end{bmatrix}. \tag{80}$$

This implies that

$$\epsilon_{1t} = P_{11} u_{1t} \tag{81}$$

$$\epsilon_{2t} = P_{21} u_{1t} + P_{22} u_{2t} \tag{82}$$

$$\epsilon_{3t} = P_{31} u_{1t} + P_{32} u_{2t} + P_{33} u_{3t}. \tag{83}$$

The way variables are ordered is thefore important: the first variable will be influenced by coincident shocks coming from other variables.

Another way of seeing it:

$$\mathbb{V}[\epsilon_{1t}] = \mathbb{V}[P_{11}u_{1t}] = P_{11}^2 \tag{84}$$

$$\mathbb{V}[\epsilon_{2t}] = \mathbb{V}[P_{21}u_{1t} + P_{22}u_{2t}] = \mathbb{V}\left[\frac{P_{21}}{P_{11}}\epsilon_{1t} + P_{22}u_{2t}\right] \tag{85}$$

This last expression shows how the variance of ϵ_{2t} can be split between what actually comes from ϵ_{1t} and what comes from its own shocks.
As $\epsilon_{1t} \perp u_{2t}$, then

$$\mathbb{V}\left[\frac{P_{21}}{P_{11}}\epsilon_{1t} + P_{22}u_{2t}\right] = \frac{P_{21}^2}{P_{11}^2} + P_{22}^2 \tag{86}$$

$\Rightarrow$ Now, we need a way to compute P. Most common way: **Choleski decomposition**.

The AR(1) case:

$$
\begin{aligned}
X_t &= \Phi_0 + \Phi_1 X_{t-1} + \sqrt{\Sigma}\epsilon_t && (87)\\
&= \Phi_0 + \Phi_1 X_{t-1} + \nu_t && (88)\\
&= \Phi_0 \sum_{i=0}^{\infty} \Phi_1^i + \sum_{i=0}^{\infty} \Phi_1^i \nu_{t-i}. && (89)\\
&= \Phi_0 \sum_{i=0}^{\infty} \Phi_1^i + \sum_{i=0}^{\infty} \Phi_1^i P u_{t-i}. && (90)
\end{aligned}
$$

The impulse response function is the function which for a given h computes

$$
h \mapsto \frac{\partial X_{t+h}}{\partial u_t} = \Phi_1^h P. \tag{91}
$$

Matrices $\Gamma_h 0 \Phi_1^h P$ correspond to **orthogonalized impulse responses**. Confidence intervals are usually built using bootstrap.

Time for an Application Model

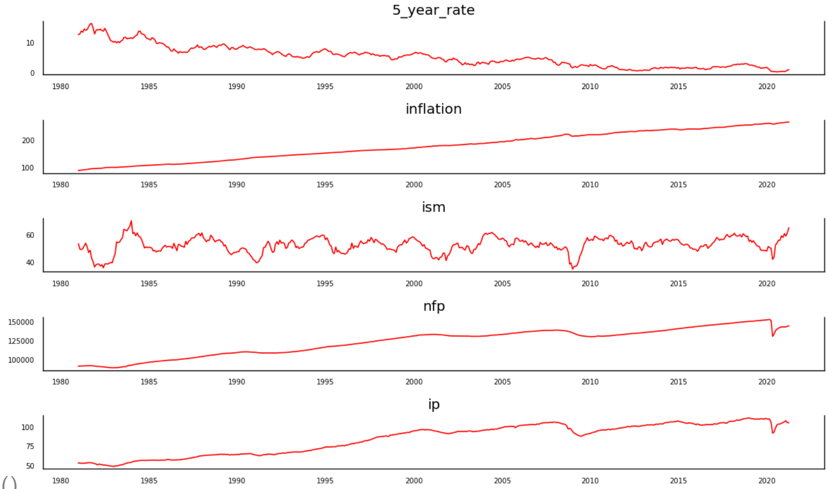
Let's use all of that with a more concrete example.

Using the following timeseries:

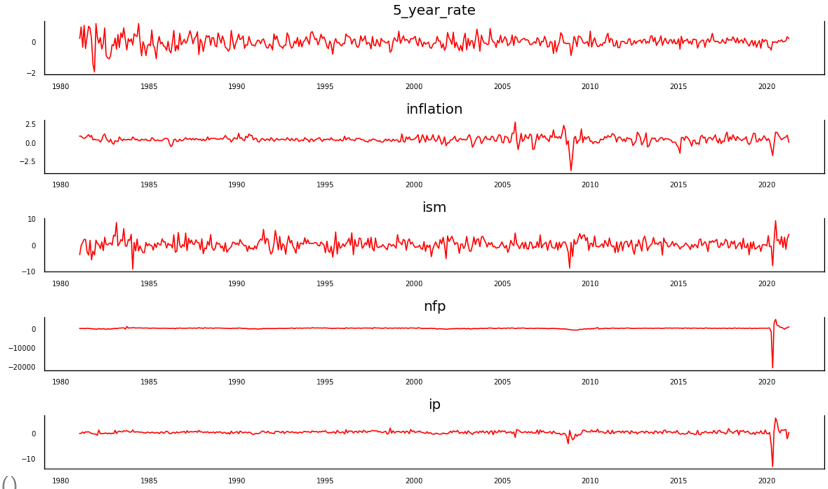
- the US 5-year rate
- the year-over-year change in the Consumer Price Index as a measure of inflation
- the US ISM, an indicator of economic conditions
- the non-farm payrolls, the number of job creation in the US
- the industrial production index.

We will use monthly time series, from December 1980 to March 2021.
Let's try creation a VAR model around these time series and try to understand what could be the impact of an inflation shock on the rest of the variables describing the US economy.

First step: Let's Take a Look at the Timeseries.



First step: Let's Take a Look at the Timeseries. With first order differences:



Second step: Granger causality test.

```
    ...:
    ...: test=grangers_causation_matrix(Macro_series, variables = Macro_series.columns)

In [90]: test
Out[90]:
              5_year_rate_x  inflation_x   ism_x   nfp_x    ip_x
5_year_rate_y        1.0000       0.0019  0.0086  0.0241  0.0453
inflation_y          0.0221       1.0000  0.0152  0.4487  0.1590
ism_y                0.0000       0.0004  1.0000  0.0000  0.0004
nfp_y                0.0258       0.0256  0.0002  1.0000  0.0000
ip_y                 0.0007       0.0011  0.0000  0.0000  1.0000

In [91]:
```

$\Rightarrow$ These series are causing each others!

Time for an Application Model

Third step: Lag Selection.

x - Object

Name	Type	Size	Value
x	tsa.vector_ar.var_model.LagOrderResults	1	LagOrderResults object
aic	int32	1	5
bic	int32	1	1
fpe	int32	1	5
hqic	int32	1	2
ics	defaultdict	4	defaultdict object of
selected_orders	dict	4	{'aic':5, 'bic':1, 'hq
summary	method	1	method object
title	str	1	VAR Order Selection (*
vecm	bool	1	False

$\Rightarrow$ No such thing as a single answer, clearly. Let's go for 2 lags.

Time for an Application Model

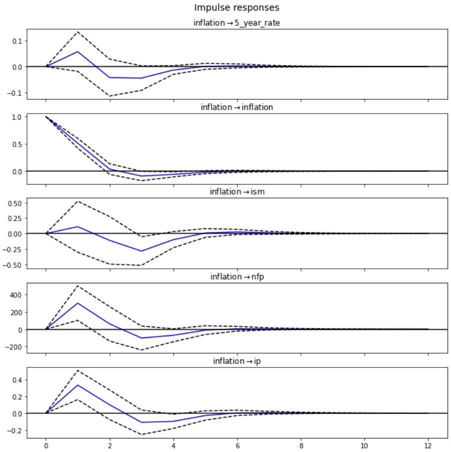
Fourth step: Parameter Estimation.

Results for equation inflation

	coefficient	std. error	t-stat	prob
const	0.262843	0.026219	10.025	0.000
L1.5_year_rate	0.080693	0.054975	1.468	0.142
L1.inflation	0.512628	0.045811	11.190	0.000
L1.ism	0.009804	0.010093	0.971	0.331
L1.nfp	-0.000078	0.000032	-2.441	0.015
L1.ip	0.079246	0.035276	2.246	0.025
L2.5_year_rate	0.065465	0.055658	1.176	0.240
L2.inflation	-0.231696	0.045487	-5.094	0.000
L2.ism	-0.005469	0.010159	-0.538	0.590
L2.nfp	-0.000132	0.000032	-4.086	0.000
L2.ip	0.142168	0.036936	3.849	0.000

$\Rightarrow$ Inflation relates positively to past industrial production.

Fifth step: Estimation of the IRFS.



$\Rightarrow$ How can we read that chart?

Variance Decomposition

We start again with the Wold decomposition of a multivariate model:

$$r_t = \varepsilon_t + \Psi_1 \varepsilon_{t-1} + \Psi_2 \varepsilon_{t-2} + ... = \Psi(L)\varepsilon_t \tag{92}$$

We are interested in the error in forecasting r_t, s periods into the future:

$$r_{t+s} - \hat{r}_{t+s/t} = \varepsilon_{t+s} + \Psi_1 \varepsilon_{t+s-1} + \Psi_2 \varepsilon_{t+s-2} + ... + \Psi_{s-1} \varepsilon_{t+1} \tag{93}$$

with

$$\hat{r}_{t+s/t} = \Psi_s \varepsilon_t + \Psi_{s+1} \varepsilon_{t-1} + \Psi_{s+2} \varepsilon_{t-2} + ... \tag{94}$$

The mean squared error of the s-period-ahead forecast is

$$MSE(\hat{r}_{t+s/t}) = E[(r_{t+s} - \hat{r}_{t+s/t})(r_{t+s} - \hat{r}_{t+s/t})'] \tag{95}$$

$$= \Sigma + \Psi_1 \Sigma \, \Psi_1^{'} + \Psi_2 \Sigma \, \Psi_2^{'} + ... + \Psi_{s-1} \Sigma \, \Psi_{s-1}^{'} \tag{96}$$

Variance Decomposition

The h-step ahead forecast error variance decomposition :

$$\theta_{ij}(h) = \sigma_{ii}^{-1} \frac{\sum_{h=0}^{h-1} (e_i' \Psi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i' \Psi_h \Sigma (\Psi_h)' e_j}, \tag{97}$$

e_i a vector of zeros but for its i^{th} element equal to 1.

Rescaled quantities:

$$\tilde{\theta}_{ij}(H) = \frac{\theta_{ij}(H)}{\sum_{j=1}^{N} \theta_{ij}(H)}, \Rightarrow \sum_{j=1}^{N} \tilde{\theta}_{ij}(H) = 1 \tag{98}$$

$\tilde{\theta}_{ij}(H)$ are "**variance shares**": measure of the share of the total forecasting error variance for series i that is explained through series j.

Variance Decomposition

- **Spillover index**: $S(H)$ measure of how much the series considered are receiving shocks that are not theirs:

$$S(H) = \frac{\sum_{i,j=1, i\neq j}^{N} \tilde{\theta}_{ij}(H)}{N} \times 100. \tag{99}$$

- **Spillover received**:

$$S_{i\leftarrow}(H) = \frac{\sum_{j=1, j\neq i}^{N} \tilde{\theta}_{ij}(H)}{\sum_{j=1}^{N} \tilde{\theta}_{ij}(H)} \times 100. \tag{100}$$

- **Spillover given**:

$$S_{i\rightarrow}(H) = \frac{\sum_{j=1, j\neq i}^{N} \tilde{\theta}_{ji}(H)}{\sum_{j=1}^{N} \tilde{\theta}_{ji}(H)} \times 100. \tag{101}$$

- **Net spillovers**:

$$S_{i}(H) = S_{i\rightarrow}(H) - S_{i\leftarrow}(H). \tag{102}$$

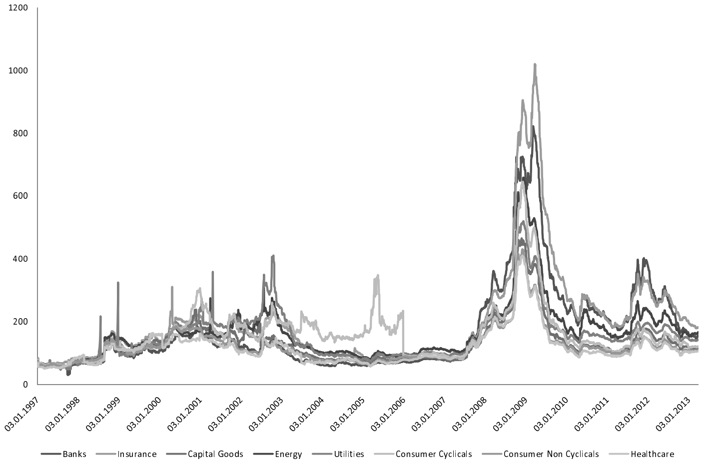
Variance Decomposition

Empirical application to credit spreads:

1. When dealing with credit portfolio, **sector exposure** is a natural partition used by credit investment managers to gauge the origin of their performance.
2. Selected sectors are supposed to **react more aggressively** to market evolutions than others: bonds issued by financial firm vs. pharmaceutical companies.
3. However sensitivity to market evolutions says little on **how shocks spread across sectors** / within which sector do turmoils usually start and end.
4. This paper provides empirical evidence that **not every sectors are net volatility spillover givers**.

Variance Decomposition n

Evolution of credit spreads by sector

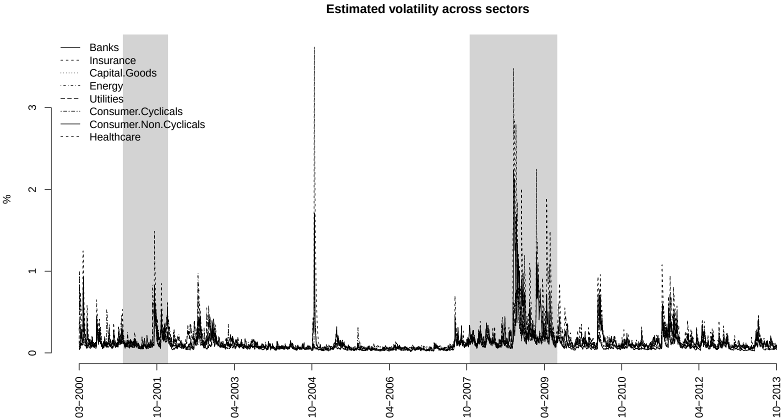


Variance Decomposition

- **Dataset**:
    - Excess returns on credit bonds, from Merill Lynch BoA.
    - US Investment Grade bonds.
    - 2001-2013.
    - By sectors: Banks, Insurance, Cap. goods, Energy, Util., Cons. cyc., Cons. non cyc., Health.
- **Based on blended volatility measures**.

Variance Decomposition

Time series estimates of each sector's volatility



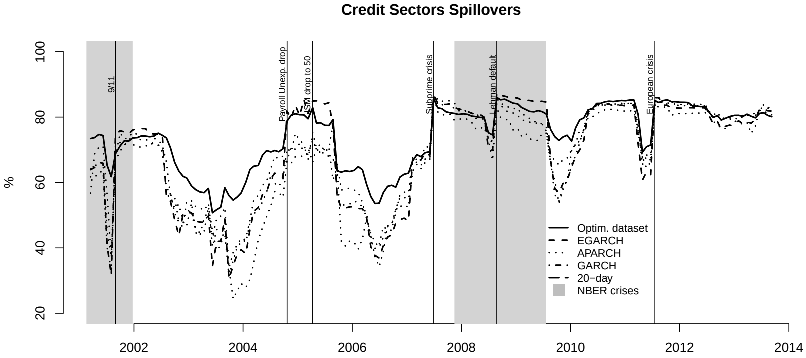
Variance Decomposition

Spillovers over the 2001-2013 period

	Banks	Ins.	Cap. Goods	Ener.	Util.	CC	CNC	Health.	Sp. Rec.
Banks	13.42	40.29	3.76	7.86	5.62	7.62	13.25	8.17	86.58
Ins.	0.47	76.77	1.17	1.05	0.74	2.65	16.79	0.36	23.23
Cap. Goods	3.09	48.94	8.68	3.92	3.52	16.5	13.19	2.16	91.32
Ener.	5.74	31.85	2.88	21.51	15.24	5.91	12.95	3.92	78.49
Util.	5.09	29.78	3.32	19.56	17.91	5.61	15.35	3.39	82.09
CC	2.85	51.04	7.67	3.98	2.95	17.69	11.62	2.19	82.31
CNC	0.67	68.68	1.27	1.37	1.26	2.5	23.75	0.5	76.25
Health.	12.38	40.08	4.02	8.11	5.59	8.99	12.01	8.81	91.19
Sp. Given	69.3	80.18	73.52	68.08	66.1	73.78	80.03	70.13	**76.43**
Net Sp.	-17.28	56.95	-17.8	-10.41	-15.99	-8.53	3.78	-21.06	

Variance Decomposition

Time series estimates of each sector's volatility



Practical Application: Inflation Shocks and Fed Rates in a VAR

Instructions

- Open the notebook `EMIF_VAR_inflation_shock_notebook_executed.ipynb`.
- Run the code cell by cell and read the comments before moving to the next block.
- Focus on four questions:
  - How do we choose the VAR lag length?
  - How do we interpret the coefficients and diagnostics?
  - What happens after a one standard deviation inflation shock?
  - Why did a model estimated at the end of 2021 fail to fully anticipate 2022?
- The exported charts and tables are used in the slides so that you can present the key messages in class.

Economic setup

We study the monthly interaction between inflation, Fed rates, and capacity utilization.

The reduced-form VAR with p lags is:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + u_t \tag{103}$$

with:

$$y_t = \begin{pmatrix} \text{Inflation}_t \\ \text{Fed Rates}_t \\ \text{Capacity Utilization Rate}_t \end{pmatrix} \tag{104}$$

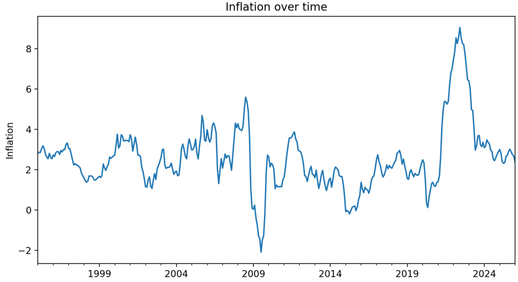
The ordering matters for the shock analysis. Inflation is placed first so that a contemporaneous inflation shock can affect the other variables within the same month.

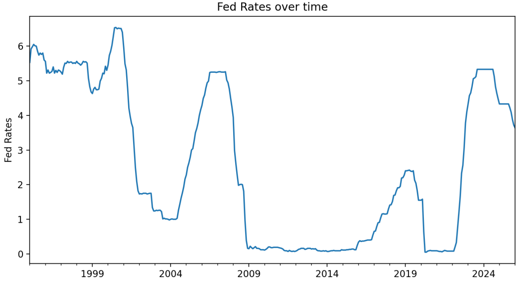
Summary statistics

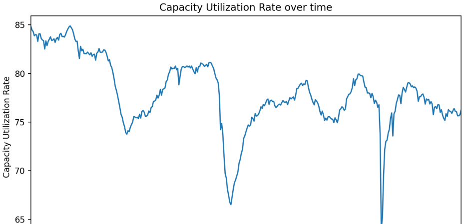
Summary statistics

	mean	std	min	max
Inflation	2.5481	1.6066	-2.0972	9.0597
Fed Rates	2.5552	2.2395	0.05	6.54
Capacity Utilization Rate	77.9678	3.6083	64.0847	84.8918

The raw data


Inflation over time
Inflation
8
6
4
2
0
−2
1999
2004
2009
2014
2019
2024


Fed Rates over time
Fed Rates
6
5
4
3
2
1
0
1999
2004
2009
2014
2019
2024


Capacity Utilization Rate over time
Capacity Utilization Rate
85
80
75
70
65

Choosing the lag length

Lag selection criteria

	AIC	BIC	HQIC	FPE
0	4.4063	4.4386	4.4192	81.9667
1	-5.8876	-5.7583	-5.8362	0.0028
2	-6.6412	-6.4149	-6.5512	0.0013
3	-6.7195	-6.3963	-6.591	0.0012
4	-6.7416	-6.3215	-6.5746	0.0012
5	-6.7122	-6.1952	-6.5067	0.0012
6	-6.7182	-6.1042	-6.4741	0.0012
7	-6.7105	-5.9995	-6.4278	0.0012
8	-6.676	-5.8681	-6.3548	0.0013
9	-6.6594	-5.7545	-6.2996	0.0013
10	-6.6406	-5.6387	-6.2423	0.0013
11	-6.6718	-5.573	-6.2349	0.0013
12	-6.6451	-5.4493	-6.1697	0.0013

Selected coefficient tables

Inflation equation coefficients

	coef	std_err	t_stat	p_value
const	-0.1491	0.5804	-0.2568	0.7973
L1.Inflation	1.4255	0.0537	26.5332	0.0
L1.Fed Rates	0.43	0.1528	2.815	0.0049
L1.Capacity Utilization Rate	0.0232	0.0275	0.8436	0.3989
L2.Inflation	-0.653	0.093	-7.019	0.0
L2.Fed Rates	-0.6866	0.2783	-2.4675	0.0136
L2.Capacity Utilization Rate	0.0137	0.0404	0.3388	0.7348
L3.Inflation	0.249	0.0928	2.6828	0.0073
L3.Fed Rates	-0.0067	0.2852	-0.0233	0.9814
L3.Capacity Utilization Rate	0.0113	0.0393	0.2879	0.7734
L4.Inflation	-0.065	0.0539	-1.2055	0.228
L4.Fed Rates	0.2616	0.1557	1.6805	0.0929
L4.Capacity Utilization Rate	-0.0447	0.0265	-1.6864	0.0917

Fed rates equation coefficients

	coef	std_err	t_stat	p_value
const	-0.2405	0.2027	-1.1864	0.2355
L1.Inflation	-0.0229	0.0188	-1.2187	0.223
L1.Fed Rates	1.5461	0.0533	28.9819	0.0
L1.Capacity Utilization Rate	0.0141	0.0096	1.4747	0.1403
L2.Inflation	0.0493	0.0325	1.5184	0.1289
L2.Fed Rates	-0.5358	0.0972	-5.5137	0.0
L2.Capacity Utilization Rate	-0.0231	0.0141	-1.6417	0.1007
L3.Inflation	-0.0682	0.0324	-2.1043	0.0354
L3.Fed Rates	0.1035	0.0996	1.039	0.2988
L3.Capacity Utilization Rate	0.0247	0.0137	1.8002	0.0718
L4.Inflation	0.0501	0.0188	2.6641	0.0077
L4.Fed Rates	-0.1265	0.0544	-2.3281	0.0199
L4.Capacity Utilization Rate	-0.0125	0.0093	-1.3467	0.1781

Identification of the inflation shock

For impulse responses, we use an orthogonalized innovation:

$$u_t = P\varepsilon_t \tag{105}$$

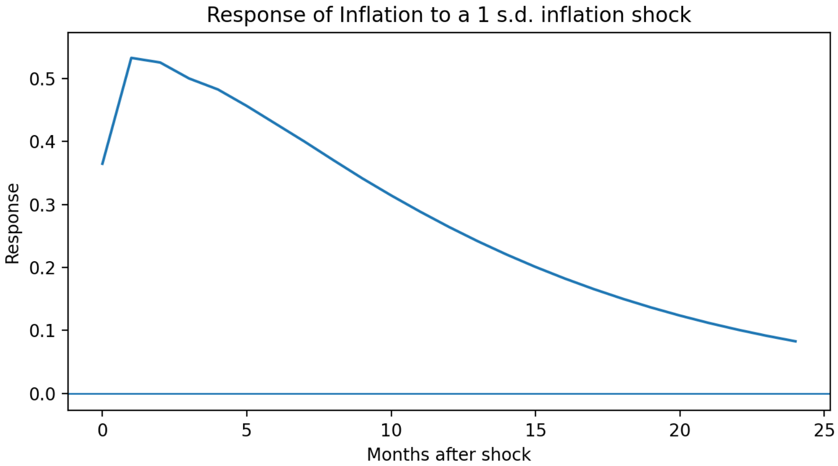
with:

$$\Sigma_u = PP' \tag{106}$$

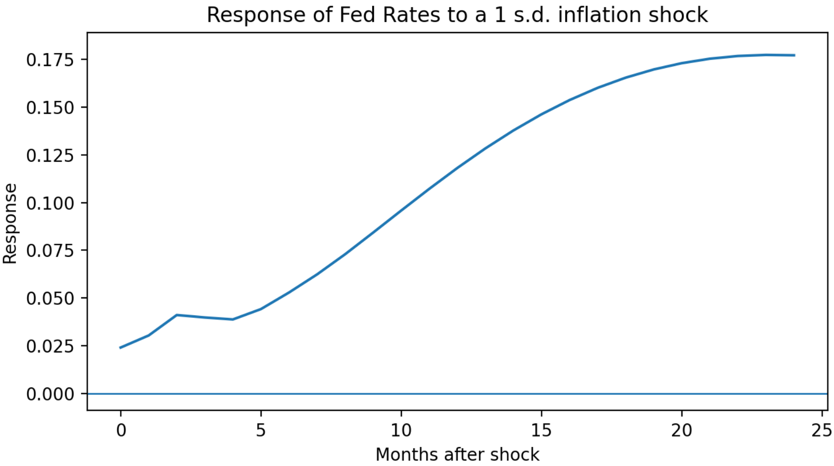
A one standard deviation inflation shock is then traced through the system. The interpretation should focus on:

- the sign of the response,
- the magnitude of the response,
- the persistence of the response.

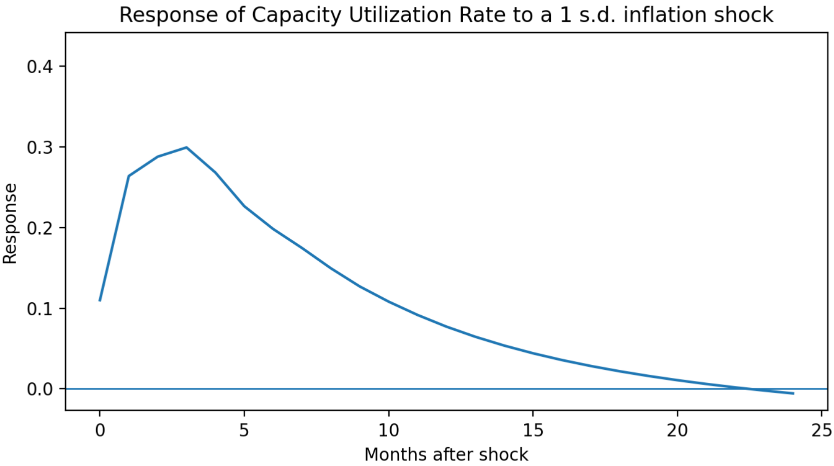
Impulse response of inflation



Impulse response of Fed rates



Impulse response of capacity utilization



Forecasting from the end of 2021

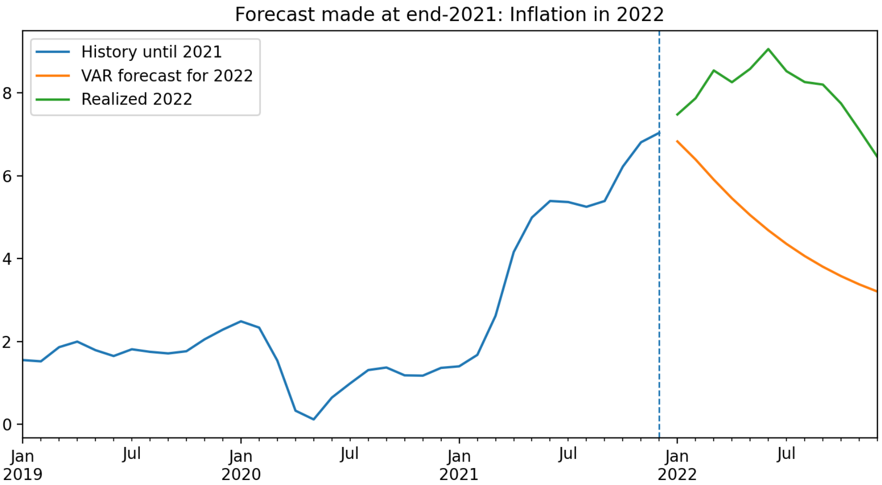
We estimate the VAR only on information available up to December 2021 and forecast the twelve months of 2022.

The key comparison is:

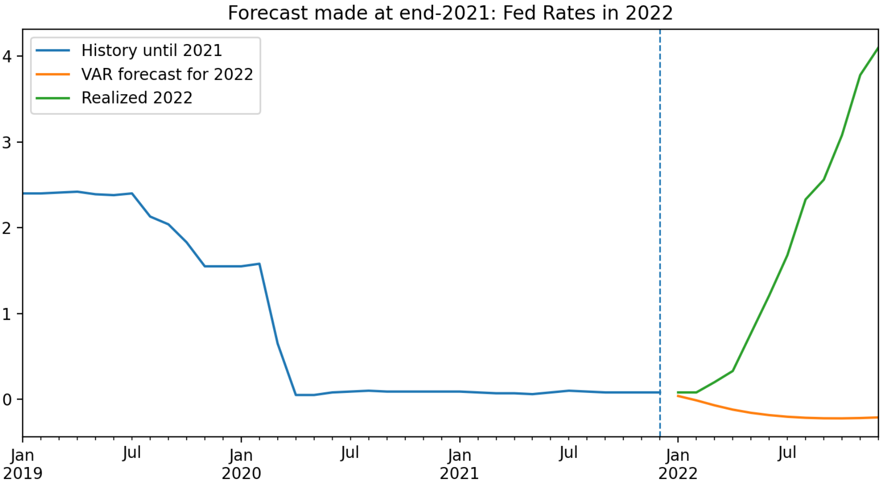
$$\text{Forecast made at end-2021} \quad \text{vs} \quad \text{Realized values in 2022} \tag{107}$$

If 2022 inflation and Fed rates are much higher than forecast, this is evidence of a large inflation shock that was not already embedded in the historical dynamics of the system.

Inflation: forecast versus realized 2022



Fed rates: forecast versus realized 2022



Forecast errors in 2022

Forecast vs realized averages in 2022

	0
Forecast average 2022 inflation	4.7253
Realized average 2022 inflation	8.0077
Forecast average 2022 Fed rates	-0.15
Realized average 2022 Fed rates	1.6833

2022 forecast RMSE

	RMSE
Inflation	3.479
Fed Rates	2.3512
Capacity Utilization Rate	1.8062

The miss in 2022 is not just a forecasting failure. It is a useful economic lesson: when inflation moves outside the range suggested by normal historical dynamics, policy rates must also adjust by more than the model had anticipated.

Takeaways

- A VAR is a convenient framework to study the joint dynamics of inflation, policy rates, and activity.
- Identification matters: the ordering of variables affects orthogonalized impulse responses.
- A one standard deviation inflation shock can be traced through the whole system with impulse responses.
- Forecast errors are economically informative: the gap between the end-2021 forecast and realized 2022 values illustrates the effect of an unusual inflation shock.
- The exercise shows both the usefulness and the limits of historical models in macro-finance.